

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}$	$\alpha\beta\chi$	
$\mathcal{K}_{0^+}^{\#1} \dagger$	$\frac{1}{2}(-Y_1 k^2 - 3 \overset{(2)}{\mathcal{K}}_3)$	0		
$\mathcal{K}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$3k^2 \overset{(4)}{\mathcal{K}}_{15}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$
			$\mathcal{K}_{1^-}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^-}^{\#2}{}_{\alpha}$
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	0	0	0
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	0	0	0
	$\mathcal{K}_{1^-}^{\#1} \dagger^{\alpha}$	0	0	$-k^2 \overset{(4)}{\mathcal{K}}_2 - \overset{(2)}{\mathcal{K}}_3$
	$\mathcal{K}_{1^-}^{\#2} \dagger^{\alpha}$	0	0	$\frac{Y_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_3}{2\sqrt{2}}$
			$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1}{}_{\alpha\beta\chi}$
			$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$Y_4 k^2$
			$\mathcal{K}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0
				0

$$\begin{array}{c}
\mathcal{T}_{0^+}^{\#1} \quad \mathcal{T}_{0^+}^{\#1 \alpha\beta\chi} \\
\mathcal{T}_{0^+}^{\#1} \dagger \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(0^+)} & 0 \\ \hline \end{array} \\
\mathcal{T}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|c|} \hline 0 & \frac{1}{3k^2 \kappa_{15}^{(4)}} \\ \hline \end{array}
\end{array}
\begin{array}{c}
\mathcal{T}_{1^+}^{\#1 \alpha\beta} \quad \mathcal{T}_{1^+}^{\#2 \alpha\beta} \quad \mathcal{T}_{1^+}^{\#1 \alpha} \quad \mathcal{T}_{1^+}^{\#2 \alpha} \\
\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta} \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha} \begin{array}{|c|c|c|c|} \hline 0 & 0 & \frac{\Upsilon_3 k^2 - \kappa_3^{(2)}}{2 \text{Det}(1^+)} & \frac{-\Upsilon_2 k^2 - 2 \kappa_3^{(2)}}{2 \sqrt{2} \text{Det}(1^+)} \\ \hline \end{array} \\
\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha} \begin{array}{|c|c|c|c|} \hline 0 & 0 & \frac{-\Upsilon_2 k^2 - 2 \kappa_3^{(2)}}{2 \sqrt{2} \text{Det}(1^+)} & \frac{-k^2 \kappa_2 - \kappa_3^{(2)}}{\text{Det}(1^+)} \\ \hline \end{array}
\end{array}
\begin{array}{c}
\mathcal{T}_{2^+}^{\#1 \alpha\beta} \quad \mathcal{T}_{2^+}^{\#1 \alpha\beta\chi} \\
\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta} \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(2^+)} & 0 \\ \hline \end{array} \\
\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$Y_1 = 3 \overset{(4)}{\kappa_1} - 4 \overset{(4)}{\kappa_{15}} + 3 \overset{(4)}{\kappa_2} + 2 \overset{(4)}{\kappa_3} \& Y_2 = 2 \overset{(4)}{\kappa_2} + \overset{(4)}{\kappa_7} \& Y_3 = 2 \overset{(4)}{\kappa_{15}} - \overset{(4)}{\kappa_2} - \overset{(4)}{\kappa_3} - \overset{(4)}{\kappa_7} \&$$

$$Y_4 = 2 \binom{(4)}{K_{15}} - \binom{(4)}{K_3} \&\& \text{Det}(0^+) = \frac{1}{2} (-k^2 (3 \binom{(4)}{K_1} - 4 \binom{(4)}{K_{15}} + 3 \binom{(4)}{K_2} + 2 \binom{(4)}{K_3}) - 3 \binom{(2)}{K_3}) \&\&$$

$$\text{Det}(1^-) = -\frac{1}{8} k^2 (k^2 (8 \binom{(4)}{\kappa_{15}} \binom{(4)}{\kappa_2} - 4 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_7}^2) + 8 \binom{(4)}{\kappa_{15}} \binom{(2)}{\kappa_3} - 4 \binom{(4)}{\kappa_3} \binom{(2)}{\kappa_3}) \&\& \text{Det}(2^+) = k^2 (2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3})$$

Lagrangian

$$\begin{aligned} & \overset{(2)}{K_3} \mathcal{K}^\alpha{}_\beta \mathcal{K}^\chi{}_{\beta\chi} + \overset{(4)}{K_1} \partial_\beta \mathcal{K}^\delta{}_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \overset{(4)}{K_2} \partial_\chi \mathcal{K}^\delta{}_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \\ & \overset{(4)}{K_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\alpha\chi} - 4 \overset{(4)}{K_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} - 4 \overset{(4)}{K_{15}} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\chi\alpha} + \overset{(4)}{K_3} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\chi\alpha} + \\ & \overset{(4)}{K_7} \partial^\chi \mathcal{K}^\alpha{}_\beta \partial_\delta \mathcal{K}^{\delta}{}_{\chi\beta} - \overset{(4)}{K_{15}} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{K_{15}} \partial_\delta \mathcal{K}^{\delta}{}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{K_{15}} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):		$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}_\delta + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta\alpha}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}_\delta + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta\beta}_\delta$	
Total # constraint(s):	11		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{1}{(-2 \frac{(4)}{\kappa_{15} + \kappa_3})^2 \frac{(2)}{\kappa_3}}$ $(6 \frac{(4)}{\kappa_{15}} \frac{(2)}{\kappa_3} - 3 \frac{(4)}{\kappa_3} \frac{(2)}{\kappa_3} + 32 \frac{(4)}{\kappa_{15}}^2 p^2 - 32 \frac{(4)}{\kappa_{15}} \frac{(4)}{\kappa_3} p^2 + 8 \frac{(4)}{\kappa_3}^2 p^2 - 36 \frac{(4)}{\kappa_{15}} \frac{(4)}{\kappa_7} p^2 + 18 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7} p^2 + 9 \frac{(4)}{\kappa_7}^2 p^2 -$ $\sqrt{(3328 \frac{(4)}{\kappa_{15}}^4 p^4 + 208 \frac{(4)}{\kappa_3}^4 p^4 + 81 \frac{(4)}{\kappa_7}^4 p^4 - 120 \frac{(4)}{\kappa_3}^3 p^2 (\frac{(2)}{\kappa_3} - 6 \frac{(4)}{\kappa_7} p^2) + 54 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7}^2 p^2 (-\frac{(2)}{\kappa_3} + 6 \frac{(4)}{\kappa_7} p^2) +$ $72 \frac{(4)}{\kappa_3}^2 (7 \frac{(2)}{\kappa_3}^2 - 3 \frac{(4)}{\kappa_7} \frac{(2)}{\kappa_3} p^2 + 11 \frac{(4)}{\kappa_7}^2 p^4) + 64 \frac{(4)}{\kappa_{15}}^3 (15 \frac{(2)}{\kappa_3} p^2 - 2 (52 \frac{(4)}{\kappa_3} + 45 \frac{(4)}{\kappa_7}) p^4) +$ $96 \frac{(4)}{\kappa_{15}}^2 (21 \frac{(2)}{\kappa_3}^2 - 3 (5 \frac{(4)}{\kappa_3} + 3 \frac{(4)}{\kappa_7}) \frac{(2)}{\kappa_3} p^2 + (52 \frac{(4)}{\kappa_3}^2 + 90 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7} + 33 \frac{(4)}{\kappa_7}^2) p^4) - 4 \frac{(4)}{\kappa_{15}} (416 \frac{(4)}{\kappa_3}^3 p^4 -$ $180 \frac{(4)}{\kappa_3}^2 p^2 (\frac{(2)}{\kappa_3} - 6 \frac{(4)}{\kappa_7} p^2) + 27 \frac{(4)}{\kappa_7}^2 p^2 (-\frac{(2)}{\kappa_3} + 6 \frac{(4)}{\kappa_7} p^2) + 72 \frac{(4)}{\kappa_3} (7 \frac{(2)}{\kappa_3}^2 - 3 \frac{(4)}{\kappa_7} \frac{(2)}{\kappa_3} p^2 + 11 \frac{(4)}{\kappa_7}^2 p^4))))$
	1	0	$\frac{1}{(-2 \frac{(4)}{\kappa_{15} + \kappa_3})^2 \frac{(2)}{\kappa_3}}$ $(6 \frac{(4)}{\kappa_{15}} \frac{(2)}{\kappa_3} - 3 \frac{(4)}{\kappa_3} \frac{(2)}{\kappa_3} + 32 \frac{(4)}{\kappa_{15}}^2 p^2 - 32 \frac{(4)}{\kappa_{15}} \frac{(4)}{\kappa_3} p^2 + 8 \frac{(4)}{\kappa_3}^2 p^2 - 36 \frac{(4)}{\kappa_{15}} \frac{(4)}{\kappa_7} p^2 + 18 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7} p^2 + 9 \frac{(4)}{\kappa_7}^2 p^2 +$ $\sqrt{(3328 \frac{(4)}{\kappa_{15}}^4 p^4 + 208 \frac{(4)}{\kappa_3}^4 p^4 + 81 \frac{(4)}{\kappa_7}^4 p^4 - 120 \frac{(4)}{\kappa_3}^3 p^2 (\frac{(2)}{\kappa_3} - 6 \frac{(4)}{\kappa_7} p^2) + 54 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7}^2 p^2 (-\frac{(2)}{\kappa_3} + 6 \frac{(4)}{\kappa_7} p^2) +$ $72 \frac{(4)}{\kappa_3}^2 (7 \frac{(2)}{\kappa_3}^2 - 3 \frac{(4)}{\kappa_7} \frac{(2)}{\kappa_3} p^2 + 11 \frac{(4)}{\kappa_7}^2 p^4) + 64 \frac{(4)}{\kappa_{15}}^3 (15 \frac{(2)}{\kappa_3} p^2 - 2 (52 \frac{(4)}{\kappa_3} + 45 \frac{(4)}{\kappa_7}) p^4) +$ $96 \frac{(4)}{\kappa_{15}}^2 (21 \frac{(2)}{\kappa_3}^2 - 3 (5 \frac{(4)}{\kappa_3} + 3 \frac{(4)}{\kappa_7}) \frac{(2)}{\kappa_3} p^2 + (52 \frac{(4)}{\kappa_3}^2 + 90 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7} + 33 \frac{(4)}{\kappa_7}^2) p^4) - 4 \frac{(4)}{\kappa_{15}} (416 \frac{(4)}{\kappa_3}^3 p^4 -$ $180 \frac{(4)}{\kappa_3}^2 p^2 (\frac{(2)}{\kappa_3} - 6 \frac{(4)}{\kappa_7} p^2) + 27 \frac{(4)}{\kappa_7}^2 p^2 (-\frac{(2)}{\kappa_3} + 6 \frac{(4)}{\kappa_7} p^2) + 72 \frac{(4)}{\kappa_3} (7 \frac{(2)}{\kappa_3}^2 - 3 \frac{(4)}{\kappa_7} \frac{(2)}{\kappa_3} p^2 + 11 \frac{(4)}{\kappa_7}^2 p^4))))$
	1	$-\frac{3 \frac{(2)}{\kappa_3}}{3 \frac{(4)}{\kappa_1} - 4 \frac{(4)}{\kappa_{15} + 3} + 3 \frac{(4)}{\kappa_2} + 2 \frac{(4)}{\kappa_3}}$	$-\frac{2}{3 \frac{(4)}{\kappa_1} - 4 \frac{(4)}{\kappa_{15} + 3} + \frac{(4)}{\kappa_2} + 2 \frac{(4)}{\kappa_3}}$
	3	$\frac{4 (-2 \frac{(4)}{\kappa_{15} + \kappa_3})^2 \frac{(2)}{\kappa_3}}{8 \frac{(4)}{\kappa_{15}} \frac{(4)}{\kappa_2} - 4 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_7}^2}$	$\frac{16 \frac{(4)}{\kappa_{15}}^2 + 4 \frac{(4)}{\kappa_3}^2 + 4 \frac{(4)}{\kappa_3} \frac{(4)}{\kappa_7} + 3 \frac{(4)}{\kappa_7}^2 - 8 \frac{(4)}{\kappa_{15}} (2 \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_7})}{(2 \frac{(4)}{\kappa_{15}} - \frac{(4)}{\kappa_3}) (8 \frac{(4)}{\kappa_{15}} \frac{(4)}{\kappa_2} - 4 \frac{(4)}{\kappa_2} \frac{(4)}{\kappa_3} + \frac{(4)}{\kappa_7}^2)}$